

BICS 2304
COMPUTER ARCHITECTURE AND ASSEMBLY LANGUAGE

PROJECT TITLE
**WEARABLE HEAT STROKE & DEHYDRATION
 PREVENTION SYSTEM**

Submission 3 : Test Cases

SECTION: 04
 LECTURER: YASIR MEHMOOD

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Test Case 1: Baseline Safe System Check

Field	Description
ID	TC-01
Title	Normal Environmental Tracking Verification
Pre-Conditions	1. MARS IDE settings configured: "Allow extended instructions" enabled. 2. Program assembled successfully with no syntax compilation breaks. 3. Hardware constants initialized: HR_threshold = 130, Temp_Threshold = 38C.
Test Steps	1. Execute the main program routine loop. 2. Enter normal physiological data at the prompts: - Temperature: 36 - Heart Rate: 80 - Accelerometer G-Force: 15 - Sustained Motion Timer: 2 3. Observe console register routing loops.
Expected Results	The application smoothly clears all priority boundary comparisons without branching into alert loops. The OLED console interface renders standard data outputs: [Status: Safe] and confirms all external actuators remain in an OFF state.

Test Case 2 : Exertional Heat Stroke Prevention Alert

Field	Description
ID	TC-02
Title	Heat Stress Index Threshold Intersection Verification
Pre-Conditions	1. The program is executing cleanly within the main execution loop frames. 2. The system is processing incoming real-time simulated sensor values.
Test Steps	1. Provide simultaneous high-stress vital signals when prompted: - Temperature: 39 (Exceeds 38C boundary limit) - Heart Rate: 140 (Exceeds 130BPM boundary limit) - Accelerometer G-Force: 10 - Sustained Motion Timer: 4 2. Press Enter to process the data payload package.
Expected Results	The system computes the compound logical AND gate matrix condition successfully. Execution intercepts lower checks and routes directly to the state_take_break block. The display prints [Status: Take a break] and sets the Haptic Motor to a continuous pulse and the Cooling Fan to active running mode.

Test Case 3 : Automated Hydration Milestone Trigger

Field	Description
ID	TC-03
Title	Sustained Active Movement Time Tracking and Timer Reset
Pre-Conditions	1. Vital signs (Temp and Heart Rate) are within baseline safe limits. 2. Hardware timer limits are initialized (s6 = 15mins).
Test Steps	1. Input values simulating steady physical work over time: - Temperature: 37 - Heart Rate: 85 - Accelerometer G-Force: 12 - Sustained Motion Timer: 15 (Matches milestone limit target) 2. Submit values via the MARS I/O terminal panel interface.
Expected Results	The program bypasses Priority 1 and 2 alert loops. It hits the time validation branch condition, executing a status update reading [Drink Water]. The Hydration LED is set to blinking, the Haptic Motor executes a brief pulse sequence, and register \$t9 resets back to 0 automatically to clear the timer loop.

Test Case 4 : Priority Layer 1 Critical Fall Emergency Override

Field	Description
ID	TC-04
Title	Accelerometer Impact Spike & Motionlessness Override Loop
Pre-Conditions	1. The system is actively polling sensor values in the infinite master monitoring loop block.
Test Steps	1. Simulate a heavy physical impact deceleration event followed by a loss of consciousness: - Temperature: 36 - Heart Rate: 75 - Accelerometer G-Force: 110 (Exceeds sudden impact threshold) - Sustained Motion Timer: 0 (Confirms absolute immobility after impact) 2. Execute data parsing cycle sequence.
Expected Results	The controller matches the high-priority Fall conditions first, jumping over all secondary physiological evaluation layers. The OLED interface displays [Status: CRITICAL], triggers simulated wireless network data streams via the SIM800L module, logs the raw GPS location matrix strings, and cleanly runs program exit code 10.

Test Case 5 : Defensive Input Boundary Validation Trap

Field	Description
ID	TC-05
Title	Out-of-Bounds Sensor Corruption Mitigation (Error Handling)
Pre-Conditions	1. Core polling procedures are active and waiting to accept user inputs.
Test Steps	1. Enter an impossible physiological data parameter to simulate a corrupted sensor wire fault: 120 at the Temperature entry line. 2. Attempt to force data submission. 3. Enter a standard recovery parameter value: 37.
Expected Results	The custom boundary validation trap catches the invalid value (120) before it passes into processing memory. The system blocks register writes, prints [ERROR]: Out of human physiological bounds! Enter 0 to 50., loops back, and keeps execution running cleanly without crashing. Entering 37 then unlocks the loop cleanly.